

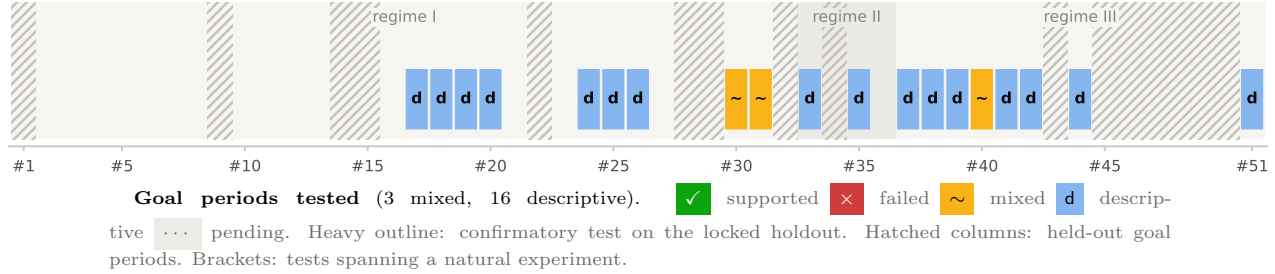
## H27: Critical slowing down warns of herding waves ahead

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2/5

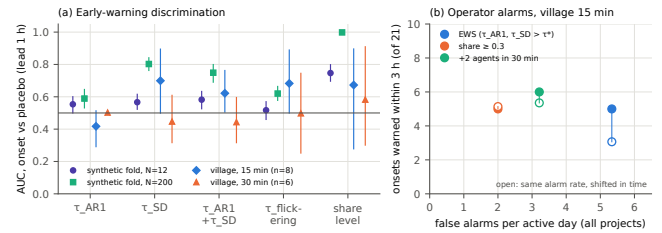
**Bottom line.** Critical slowing down does not warn of herding in the AI Village: pile-ons in attention and in commits are fast, usually marked by a link, and too few agents are watched for slowing down to show.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



### WHAT HAPPENED

A pre-registered rule on project labels defined herding onsets: a project's share of labeled agents jumps from  $\leq 0.25$  to  $\geq 0.5$  within an hour and stays there. It found 21 onsets in 14 non-holdout weeks at 15-min resolution, mostly in shared-artifact weeks. Early-warning trends (rolling lag-1 autocorrelation and SD over the previous 6 h) were compared with matched placebo segments; 8 onsets were evaluable. The composite separated them weakly at 15 min (AUC 0.62) and not at 30 min; autocorrelation never rose (0.41), so this is not critical slowing down. **Round 1b** (deterministic labels, work ledger) reproduces all of it: same onsets, same AUC. The frozen alarm now beats a time-shifted copy at  $p = 0.038$  (was 0.07), but a naive share alarm matches its 5/21 hits with fewer alarms. Work-commit onsets coincide with attention onsets, and #40's pile-on ran in minutes after a link. Holdout not run.



**Observables.** (a) AUC of each early-warning trend, onset vs placebo segments, at 1 h lead. Series: a synthetic fold at village size (purple), a clean  $N = 200$  fold (green), and the village at 15 min (blue, 8 onsets) and 30 min (orange, 6). Autocorrelation never discriminates; variance does only at 15 min, with wide CIs. (b) Onsets warned within 3 h against false alarms per active day. Open circles: the same alarms shifted in time. No alarm beats its shifted copy.

### SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator this is a useful negative, now checked on cleaner data and in commits. Generic early-warning indicators (rising autocorrelation and variance) do not forecast pile-ons in a 10–15-agent LLM swarm. Pile-ons are fast pushes: a link posted in chat, then half the room within minutes (#40: 4 min to half the room's attention, 19 min to half its commits). A 6-hour trend window cannot see that, and at  $N \approx 12$  a true fold would be barely detectable anyway. So don't build a critical-slowness dashboard for swarms this size. Watch announcements and shares instead, and size any response to minutes, not hours. The transferable lesson: test alarm timing against a rate-matched shift; the round-1 vs 1b  $p$  (0.07 vs 0.04) shows how thin such timing signals are.

### PHYSICAL MODEL

Models 10 (kinetic mean-field Potts) and 01 (mean-field susceptibility).  $\sigma_i$ : agent  $i$ 's project in a window ( $0 =$  rare projects, uncoupled).  $k_a^{(-i)}$ : other agents on  $a$ ;  $h_a(t)$ : a slowly drifting popularity field;  $\beta J$ : uniform herding coupling;  $\gamma$ : reconsideration rate;  $x_a$ : share on  $a$ . Near a fold of  $x_A$  the leading eigenvalue  $\lambda \rightarrow 0$ , so the lag-1 autocorrelation  $\rho_1$  and the variance rise (critical slowing down). Rivals (field steps, noise-induced flips) give no warning.

$$P(\sigma_i \rightarrow a) \propto e^{h_a(t) + \beta J k_a^{(-i)} / (N-1)} (a \geq 1), \quad \propto 1 (a = 0)$$

$$\dot{x}_a = \gamma [\text{softmax}_a(h(t) + \beta J x) - x_a]$$

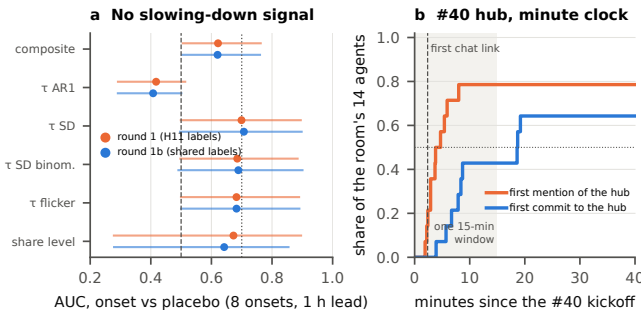
$$\rho_1 = e^{-\gamma |\lambda| \Delta t} \rightarrow 1, \quad \text{Var } x_A \propto 1/(N|\lambda|) \quad (\lambda \rightarrow 0)$$

$$\text{alarm: } \tau_K(\hat{\rho}_1) > \tau^* \wedge \tau_K(\hat{\sigma}) > \tau^*, \quad \tau^* = 0.54$$

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## RESULTS VS PREDICTIONS

| Prediction            | Observed (round 1 → 1b)                                    | Verdict |
|-----------------------|------------------------------------------------------------|---------|
| P0: 10–25 onsets      | 21 → 21; 8 evaluable                                       | yes     |
| P1: $AUC \geq 0.70$   | 0.62 → 0.62 [0.50, 0.76]; 30 min<br>0.44 → 0.50            | mixed   |
| P2: $SD > AR1$        | 0.70/0.42 → 0.71/0.41                                      | partly  |
| P3: no operator value | 5/21 hits; shift $p$ 0.07 → 0.038; mostly level alarm 5/21 |         |
| P4: synthetic         | $N=200$ : 0.75; $N=12$ : 0.58                              | 2 of 4  |
| R1b-1 replication     | same onsets, same verdicts                                 | yes     |
| R1b-2 work onsets     | 8 vs 14; lag 0 (3) or +1 (1) window                        | mostly  |
| R1b-3 links           | 11/21; 9 to known projects, 2 first links                  | yes     |
| #40 minute clock      | $t_{50}$ 1.5 min after link; work 15 min later             | 2 of 3  |



**Second figure.** (a) AUC of each trend, onset vs placebo, at 1 h lead: round 1 (orange) vs round 1b shared labels (blue). Nothing moves; autocorrelation stays below 0.5. (b) #40 at the minute clock: share of the room with a first mention of the hub (orange) and a first commit to it (blue) after the kickoff. Half the room is in within one 15-min window (shaded).

## GOAL PERIODS IN DETAIL

G17 d 1b: descriptive (unchanged)  
 G18 d 1b: descriptive (unchanged)  
 G19 d 1b: descriptive (unchanged)  
 G20 d 1b: descriptive (unchanged)  
 G24 d 1b: descriptive (unchanged)  
 G25 d 1b: descriptive (unchanged)  
 G26 d 1b: descriptive (unchanged)  
 G30 ~ 1b: mixed (AUC 0.55, 2/4 hit; unchanged)  
 G31 ~ 1b: mixed (native: work onsets coincide with attention)  
 G33 d 1b: descriptive (unchanged)  
 +9 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 0 | G ground truth    | 1 |
| C adequacy             | 0 | H comparative     | 0 |
| D unfitted predictions | 0 | I transfer        | 0 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

21 onsets, 8 evaluable, 4 in work: per-week AUCs rest on 1–2 onsets. Many indicators, leads, window lengths and two onset rules; only the 15-min composite at 1 h was primary. The alarm-vs-shift  $p$  crosses 0.05 between null draws (0.038–0.07): treat it as borderline. The step rule misses slow rises; the slow-rise variant gives 0.74 at 15 min. The link association is post hoc; most linked onsets are bursts of known projects (H28). Work labels are public agent commits (dense from #30). #51 supplies most false alarms.

## ROUND 2

- **Essence.** Can an operator see a pile-on coming early enough to act, and from what signal?
- **H27-R1** Model onsets as announcement-triggered cascades: hazard of a majority pile-on after a link post, by poster, room size and the project's prior share; joint with H28.
- **H27-R2** Use a finer clock: per-turn project states from actions (minutes, not 15-min windows) so 6 h of history is not needed and the first 1.5 days become evaluable.
- **H27-R3** Test the one regime where the theory says warnings can work: long, large swarms with slowly drifting incentives (#51 roles, if any shared project ever rises), with the slow-rise onset rule.

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